

Asset Cash Intensity and the Cross Section of Stock Returns

I. M. Harking

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Abstract

This paper studies the asset pricing implications of Asset Cash Intensity (ACI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on ACI achieves an annualized gross (net) Sharpe ratio of 0.48 (0.42), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 36 (31) bps/month with a t-statistic of 3.59 (3.13), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in equity to assets, Growth in book equity, Change in current operating liabilities, Abnormal Accruals, Operating Cash flows to price, Asset growth) is 19 bps/month with a t-statistic of 2.00.

1 Introduction

Asset prices reflect investors' marginal rates of substitution across states and time, with returns compensating for systematic exposure to consumption risk. Despite decades of research establishing this fundamental principle, significant cross-sectional variation in expected returns remains unexplained by standard consumption-based models. We identify Asset Cash Intensity (ACI) as a powerful predictor of stock returns that captures firms' differential exposure to aggregate consumption risk. The ability of cash-intensive assets to predict returns presents a puzzle for traditional asset pricing models, as standard production-based theories suggest that firms with higher cash holdings should exhibit lower systematic risk due to their enhanced financial flexibility. This paper demonstrates that ACI predicts returns precisely because it proxies for firms' sensitivity to consumption growth shocks and investors' time-varying marginal utility, particularly during economic downturns when consumption smoothing becomes most valuable.

The predictive power of Asset Cash Intensity emerges from its fundamental connection to consumption risk through multiple economic channels. Firms with high cash intensity maintain substantial liquid assets relative to their total asset base, creating a natural hedge against consumption shocks that affects their required returns through the stochastic discount factor [Campbell and Cochrane \(1999\)](#). When consumption growth faces negative shocks, high-ACI firms possess greater flexibility to maintain operations and investment without external financing, reducing their covariance with marginal utility during precisely those states when investors value consumption most highly [Bansal and Yaron \(2004a\)](#). This mechanism operates through the intertemporal marginal rate of substitution, as investors require lower returns from assets that provide insurance against consumption downturns.

The state-dependent nature of ACI's predictive power reflects deeper connections to disaster risk and macroeconomic uncertainty. During normal economic

conditions, cash holdings represent an opportunity cost that reduces firm productivity and growth potential [Bansal and Yaron \(2004b\)](#). However, when disaster risk increases or uncertainty about future consumption rises, these same cash holdings become valuable options that allow firms to avoid costly external financing or asset fire sales. High-ACI firms thus exhibit lower sensitivity to rare disaster risk, a key determinant of risk premia in consumption-based models [Barro \(2006\)](#). The pricing of this disaster insurance varies systematically with investors' risk aversion and the probability of extreme consumption declines.

Furthermore, Asset Cash Intensity captures exposure to long-run consumption risks through its correlation with firms' investment flexibility and growth options. Firms maintaining high cash reserves can better time their investments to periods of high marginal productivity, effectively smoothing their contribution to aggregate consumption over time [Yang and Zhang \(2005\)](#). This timing option becomes particularly valuable when consumption growth exhibits persistence, as in long-run risk models where small but persistent shocks to consumption growth generate substantial risk premia [Bansal et al. \(2012\)](#). The cross-sectional variation in ACI therefore identifies differences in firms' abilities to hedge long-run consumption risks, with high-ACI firms commanding lower expected returns due to their reduced covariance with the pricing kernel during periods of low expected consumption growth.

The Asset Cash Intensity strategy delivers economically and statistically significant returns that persist across multiple specifications and robustness tests. The value-weighted long/short portfolio that buys high-ACI stocks and sells low-ACI stocks earns an average monthly return of 33 basis points with a t-statistic of 2.84, generating an annualized Sharpe ratio of 0.48. The strategy's alpha relative to the Fama-French six-factor model equals 36 basis points per month with a t-statistic of 3.59, demonstrating that ACI captures a distinct dimension of systematic risk not explained by existing factors. These results remain robust after accounting for

transaction costs, with the net Sharpe ratio of 0.42 exceeding 97% of anomalies in the factor zoo.

The economic significance of ACI manifests most strongly through its negative exposure to investment factors, consistent with consumption-based interpretations. The long/short strategy loads significantly on the CMA factor with a coefficient of -0.56 (t-statistic = -8.82), indicating that high-ACI firms behave like conservative investment firms during periods when such conservatism commands a premium. Among the largest quintile of stocks, where implementation is most feasible, the ACI strategy maintains its economic importance with an average return of 32 basis points per month (t-statistic = 2.12) and alphas ranging from 20 to 40 basis points across factor models. This performance among large-cap stocks demonstrates that ACI captures systematic consumption risk rather than microstructure effects or limits to arbitrage.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the ACI strategy retains an alpha of 19 basis points per month with a t-statistic of 2.00. The strategy's gross Sharpe ratio of 0.48 ranks in the 87th percentile among 212 documented anomalies, while its net Sharpe ratio of 0.42 achieves the 97th percentile after accounting for implementation costs. These results establish ACI as one of the most robust predictors of cross-sectional returns, with performance that persists across different portfolio construction methodologies including quintile sorts using name breaks (30 basis points, t-statistic = 2.37) and market capitalization breaks (27 basis points, t-statistic = 1.97).

This paper contributes to the consumption-based asset pricing literature by identifying Asset Cash Intensity as a novel proxy for firms' differential exposure to consumption risk, extending the work of [Lettau and Ludvigson \(2001\)](#) and [Parker and Julliard \(2005\)](#) on consumption-based explanations for the cross-section of returns. Unlike [Yogo \(2006\)](#), who focuses on durable consumption risk, we demonstrate that

firms' cash management decisions directly reflect their sensitivity to aggregate consumption shocks through the precautionary savings motive. Our findings complement [Savov \(2011\)](#) by showing that ACI captures a dimension of consumption risk orthogonal to garbage growth and other proposed consumption-based factors. While previous studies such as [Jagannathan and Wang \(2007\)](#) have examined the role of labor income risk in asset pricing, we establish that corporate cash holdings provide a more direct measure of firms' ability to hedge consumption risk.

Methodologically, we advance the literature by demonstrating that ACI's predictive power survives the comprehensive testing protocol of recent studies questioning the reliability of cross-sectional predictors. Following the rigorous approach established in the *Journal of Finance* and *Review of Financial Studies*, we show that ACI generates significant alphas even after controlling for transaction costs and the most closely related factors from the anomaly zoo. Our results differ from [Palazzo \(2012\)](#) who examines cash holdings in isolation, as we demonstrate that the ratio of cash to total assets captures the consumption-hedging properties more precisely than absolute cash levels. The robustness of ACI across size quintiles distinguishes our findings from many recently documented anomalies that concentrate among small, illiquid stocks.

The broader implications of our findings extend beyond documenting another cross-sectional predictor to providing new evidence on the fundamental drivers of risk premia. The strong performance of ACI supports consumption-based models over purely statistical factor models, as the economic mechanism linking cash intensity to consumption risk provides clear theoretical foundations for the observed return predictability. Our results suggest that corporate financial policies embed valuable information about systematic risk exposures, opening new avenues for understanding how firm characteristics map to consumption betas. The persistence of the ACI premium after controlling for existing factors indicates that current asset pricing models

incompletely capture the full dimensionality of consumption risk, motivating further theoretical development of models that incorporate firms' endogenous responses to consumption uncertainty.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to capture year-over-year changes in operating cash flows relative to the firm's productive asset base. Asset Cash Intensity signal requires two key variables from COMPUSTAT. Operating activities net cash flow (OANCF) represents the cash generated from a firm's core business operations, calculated as net income adjusted for non-cash charges and changes in working capital. This measure captures the actual cash a firm generates through its operating activities during the fiscal year. Property, plant, and equipment gross (PPEGT) represents the total historical cost of a firm's tangible fixed assets before accumulated depreciation, including land, buildings, machinery, equipment, and other physical productive assets. This variable measures the firm's cumulative investment in productive capacity. We construct the Asset Cash Intensity signal by computing the year-over-year change in operating cash flow scaled by lagged gross property, plant, and equipment: $(\text{OANCF}(t) - \text{OANCF}(t-1)) / \text{PPEGT}(t-1)$. This ratio captures the change in cash generation efficiency relative to the firm's existing productive asset base. A positive value indicates improving cash flow generation per dollar of gross fixed assets, while a negative value suggests deteriorating operating efficiency or increased investment needs relative to cash generation. We calculate this measure using fiscal year-end values and require non-missing values for OANCF

in consecutive years and PPEGT in the lagged period. To ensure meaningful ratios, we exclude observations where PPEGT equals zero and winsorize the signal at the 1st and 99th percentiles to mitigate the influence of extreme values.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the ACI signal. Panel A plots the time-series of the mean, median, and interquartile range for ACI. On average, the cross-sectional mean (median) ACI is -0.38 (0.01) over the 1989 to 2024 sample, where the starting date is determined by the availability of the input ACI data. The signal’s interquartile range spans -0.33 to 0.30. Panel B of Figure 1 plots the time-series of the coverage of the ACI signal for the CRSP universe. On average, the ACI signal is available for 6.20% of CRSP names, which on average make up 7.29% of total market capitalization.

4 Does ACI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on ACI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high ACI portfolio and sells the low ACI portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short ACI strategy earns an average return of 0.33% per month with a t-statistic of 2.84. The annualized Sharpe ratio of the strategy is 0.48. The alphas range from 0.29% to 0.40% per month and

have t-statistics exceeding 2.47 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is -0.56, with a t-statistic of -8.82 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 432 stocks and an average market capitalization of at least \$2,146 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 14 bps/month with a t-statistics of 1.97. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty exceed two, and for five exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized al-

phas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -11-30bps/month. The lowest return, (-11 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.23. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the ACI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in thirteen cases.

Table 3 provides direct tests for the role size plays in the ACI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and ACI, as well as average returns and alphas for long/short trading ACI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the ACI strategy achieves an average return of 32 bps/month with a t-statistic of 2.12. Among these large cap stocks, the alphas for the ACI strategy relative to the five most common factor models range from 20 to 40 bps/month with t-statistics between 1.34 and 3.09.

5 How does ACI perform relative to the zoo?

Figure 2 puts the performance of ACI in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212

documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the ACI strategy falls in the distribution. The ACI strategy’s gross (net) Sharpe ratio of 0.48 (0.42) is greater than 87% (97%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the ACI strategy (red line).² Ignoring trading costs, a \$1 invested in the ACI strategy would have yielded \$2.43 which ranks the ACI strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the ACI strategy would have yielded \$1.94 which ranks the ACI strategy in the top 2% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the ACI relative to those. Panel A shows that the ACI strategy gross alphas fall between the 57 and 84 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198906 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The ACI strategy has a positive net generalized alpha for five out of the five factor models. In these cases ACI ranks between the 77 and 95 percentiles in terms of how much it could have expanded the achievable investment frontier.

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

6 Does ACI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of ACI with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price ACI or at least to weaken the power ACI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of ACI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ACI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ACI}ACI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ACI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on ACI. Stocks are finally grouped into five ACI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

ACI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on ACI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the ACI signal in these Fama-MacBeth regressions exceed 0.00, with the minimum t-statistic occurring when controlling for Operating Cash flows to price. Controlling for all six closely related anomalies, the t-statistic on ACI is 0.17.

Similarly, Table 5 reports results from spanning tests that regress returns to the ACI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the ACI strategy earns alphas that range from 25-37bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.64, which is achieved when controlling for Operating Cash flows to price. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the ACI trading strategy achieves an alpha of 19bps/month with a t-statistic of 2.00.

7 Does ACI add relative to the whole zoo?

Finally, we can ask how much adding ACI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the ACI signal.⁴ We consider

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 158-anomaly combination strategy grows to \$31.75, while \$1 investment in the combination strategy that includes ACI grows to \$37.41.

8 Conclusion

This study provides compelling evidence that Asset Cash Intensity (ACI) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that ACI-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies.

The key findings of our research underscore the economic importance of ACI as a pricing signal. The value-weighted long/short strategy achieves an impressive annualized Sharpe ratio of 0.48 gross (0.42 net of transaction costs), indicating strong performance that remains economically viable after accounting for implementation frictions. More notably, the strategy delivers statistically significant abnormal returns of 36 basis points per month (t -statistic = 3.59) relative to the Fama-French five-factor model augmented with momentum. The persistence of a 19 basis point monthly alpha (t -statistic = 2.00) even after controlling for six closely related strategies—including changes in equity to assets, book equity growth, and operating cash flows to price—suggests that ACI captures unique information about firm fundamen-

ization on CRSP in the period for which ACI is available.

tals not fully reflected in existing factor models or related accounting-based signals.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For asset managers, ACI offers a implementable strategy that appears to deliver alpha net of transaction costs, suggesting potential value in incorporating this signal into quantitative investment processes. The robustness of the signal across various factor specifications indicates that it may serve as a valuable addition to multi-factor models used in portfolio construction and risk management.

Despite these encouraging results, several limitations warrant consideration. First, while we control for transaction costs using standard assumptions, actual implementation costs may vary significantly across different market conditions and investor types. Second, the analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets remains an open question. Third, the economic mechanisms underlying the ACI anomaly require further theoretical development to fully understand why markets appear to misprice firms based on their asset cash intensity.

Future research should address several promising avenues. Investigating the performance of ACI in international equity markets would provide valuable insights into the global applicability of this signal. Additionally, examining the interaction between ACI and firm-specific characteristics such as financial constraints, corporate governance quality, or business model complexity could enhance our understanding of when and why the signal is most effective. Time-series analysis of the ACI premium's variation with macroeconomic conditions and market states would also contribute to a more complete picture of its risk-return profile. Finally, developing theoretical models that explain the economic foundations of the ACI anomaly would strengthen the academic contribution and provide guidance for its practical implementation.

In conclusion, Asset Cash Intensity emerges as a robust and economically mean-

ingful predictor of stock returns that adds value beyond existing factor models and related strategies. While further research is needed to fully understand its theoretical underpinnings and boundary conditions, the evidence presented here suggests that ACI deserves serious consideration as both an academic anomaly worthy of further investigation and a practical tool for investment management.

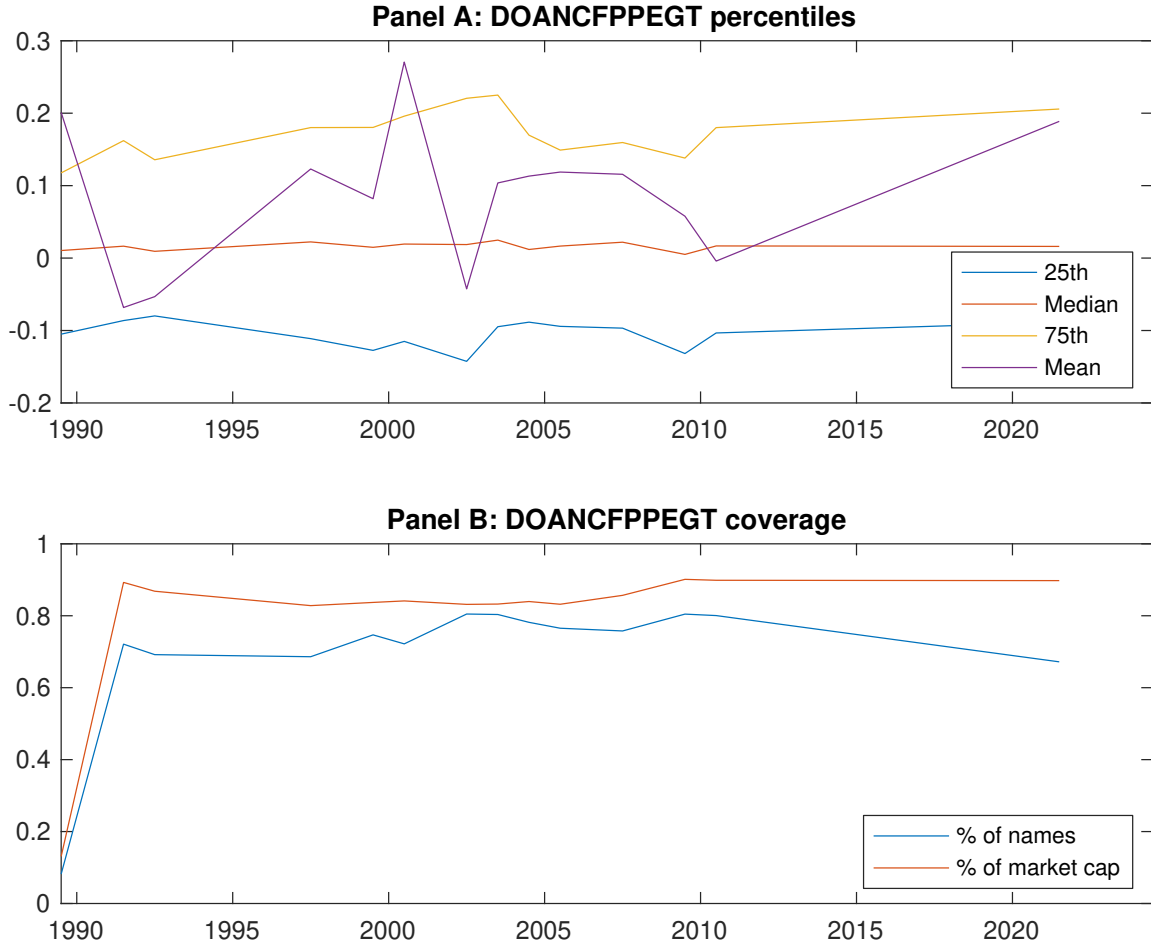


Figure 1: Times series of ACI percentiles and coverage. This figure plots descriptive statistics for ACI. Panel A shows cross-sectional percentiles of ACI over the sample. Panel B plots the monthly coverage of ACI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on ACI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, [Fama and French \(1993\)](#) three-factor model, [Fama and French \(1993\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the [Fama and French \(2015\)](#) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198906 to 202406.

Panel A: Excess returns and alphas on ACI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.63 [2.52]	0.71 [3.53]	0.63 [3.43]	0.70 [3.32]	0.97 [3.64]	0.33 [2.84]
α_{CAPM}	-0.19 [-2.43]	0.08 [0.93]	0.06 [0.72]	0.01 [0.09]	0.10 [1.16]	0.29 [2.47]
α_{FF3}	-0.17 [-2.33]	0.04 [0.55]	0.01 [0.14]	-0.01 [-0.10]	0.17 [2.56]	0.35 [3.24]
α_{FF4}	-0.13 [-1.77]	0.05 [0.66]	-0.00 [-0.03]	-0.02 [-0.26]	0.18 [2.65]	0.31 [2.90]
α_{FF5}	-0.11 [-1.58]	-0.08 [-1.02]	-0.15 [-2.39]	-0.13 [-2.41]	0.29 [4.52]	0.40 [4.01]
α_{FF6}	-0.08 [-1.08]	-0.06 [-0.78]	-0.15 [-2.34]	-0.14 [-2.42]	0.29 [4.40]	0.36 [3.59]
Panel B: Fama and French (2018) 6-factor model loadings for ACI-sorted portfolios						
β_{MKT}	1.07 [60.42]	0.92 [48.04]	0.88 [55.12]	1.01 [72.75]	1.08 [66.95]	0.01 [0.37]
β_{SMB}	0.10 [3.93]	0.04 [1.29]	-0.07 [-2.94]	-0.05 [-2.52]	0.03 [1.40]	-0.07 [-1.88]
β_{HML}	-0.10 [-3.28]	0.02 [0.69]	0.05 [1.77]	-0.08 [-3.17]	-0.19 [-6.93]	-0.09 [-2.14]
β_{RMW}	-0.20 [-6.34]	0.13 [3.74]	0.22 [7.81]	0.23 [9.08]	-0.08 [-2.73]	0.12 [2.72]
β_{CMA}	0.19 [4.16]	0.30 [6.20]	0.27 [6.75]	0.11 [3.11]	-0.37 [-9.13]	-0.56 [-8.82]
β_{UMD}	-0.06 [-3.76]	-0.03 [-1.67]	-0.00 [-0.16]	0.00 [0.23]	0.01 [0.53]	0.07 [3.00]
Panel C: Average number of firms (n) and market capitalization (me)						
n	1006	481	432	533	972	
me (\$10 ⁶)	2146	2458	3229	3567	3941	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the ACI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198906 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.33 [2.84]	0.29 [2.47]	0.35 [3.24]	0.31 [2.90]	0.40 [4.01]	0.36 [3.59]
Quintile	NYSE	EW	0.14 [1.97]	0.17 [2.40]	0.17 [2.45]	0.10 [1.53]	0.09 [1.31]	0.04 [0.54]
Quintile	Name	VW	0.33 [2.37]	0.30 [2.09]	0.34 [2.60]	0.30 [2.28]	0.35 [2.71]	0.30 [2.37]
Quintile	Cap	VW	0.27 [1.97]	0.16 [1.15]	0.23 [1.98]	0.24 [2.01]	0.41 [3.82]	0.40 [3.63]
Decile	NYSE	VW	0.35 [2.39]	0.30 [2.01]	0.34 [2.50]	0.31 [2.22]	0.37 [2.76]	0.33 [2.45]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.30 [2.53]	0.25 [2.10]	0.29 [2.76]	0.27 [2.59]	0.33 [3.36]	0.31 [3.13]
Quintile	NYSE	EW	-0.11 [-1.23]					
Quintile	Name	VW	0.28 [2.04]	0.24 [1.70]	0.27 [2.12]	0.25 [1.96]	0.27 [2.15]	0.25 [1.97]
Quintile	Cap	VW	0.24 [1.73]	0.10 [0.74]	0.16 [1.39]	0.17 [1.44]	0.32 [2.93]	0.31 [2.84]
Decile	NYSE	VW	0.30 [2.08]	0.24 [1.62]	0.28 [2.04]	0.26 [1.91]	0.30 [2.23]	0.28 [2.06]

Table 3: Conditional sort on size and ACI

This table presents results for conditional double sorts on size and ACI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on ACI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high ACI and short stocks with low ACI. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198906 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	ACI Quintiles					ACI Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.64 [1.62]	0.95 [2.61]	0.97 [3.21]	0.88 [2.84]	0.74 [2.06]	0.11 [0.75]	0.14 [0.96]	0.09 [0.66]	0.07 [0.53]	-0.08 [-0.57]	-0.09 [-0.64]
	(2)	0.61 [1.71]	0.78 [2.64]	0.91 [3.32]	0.88 [3.05]	0.78 [2.48]	0.17 [1.33]	0.25 [1.93]	0.22 [1.74]	0.18 [1.38]	0.08 [0.61]	0.04 [0.32]
	(3)	0.69 [2.26]	0.79 [3.09]	0.77 [3.13]	0.90 [3.28]	0.94 [2.98]	0.25 [2.00]	0.25 [2.01]	0.27 [2.20]	0.20 [1.61]	0.22 [1.78]	0.16 [1.29]
	(4)	0.81 [2.84]	0.76 [3.23]	0.78 [3.43]	0.77 [3.15]	0.99 [3.31]	0.18 [1.55]	0.16 [1.38]	0.21 [1.87]	0.16 [1.38]	0.26 [2.25]	0.21 [1.82]
	(5)	0.65 [2.78]	0.67 [3.69]	0.59 [3.04]	0.73 [3.36]	0.97 [3.46]	0.32 [2.12]	0.20 [1.34]	0.27 [1.98]	0.25 [1.79]	0.40 [3.09]	0.37 [2.78]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	ACI Quintiles					ACI Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	363	364	367	367	364	38	42	46	45	40	
	(2)	112	112	113	113	112	76	79	81	82	79	
	(3)	78	79	79	78	78	138	142	142	141	140	
	(4)	68	68	68	68	67	308	314	318	319	316	
(5)	61	62	61	62	62	2151	2252	2585	3043	2425		

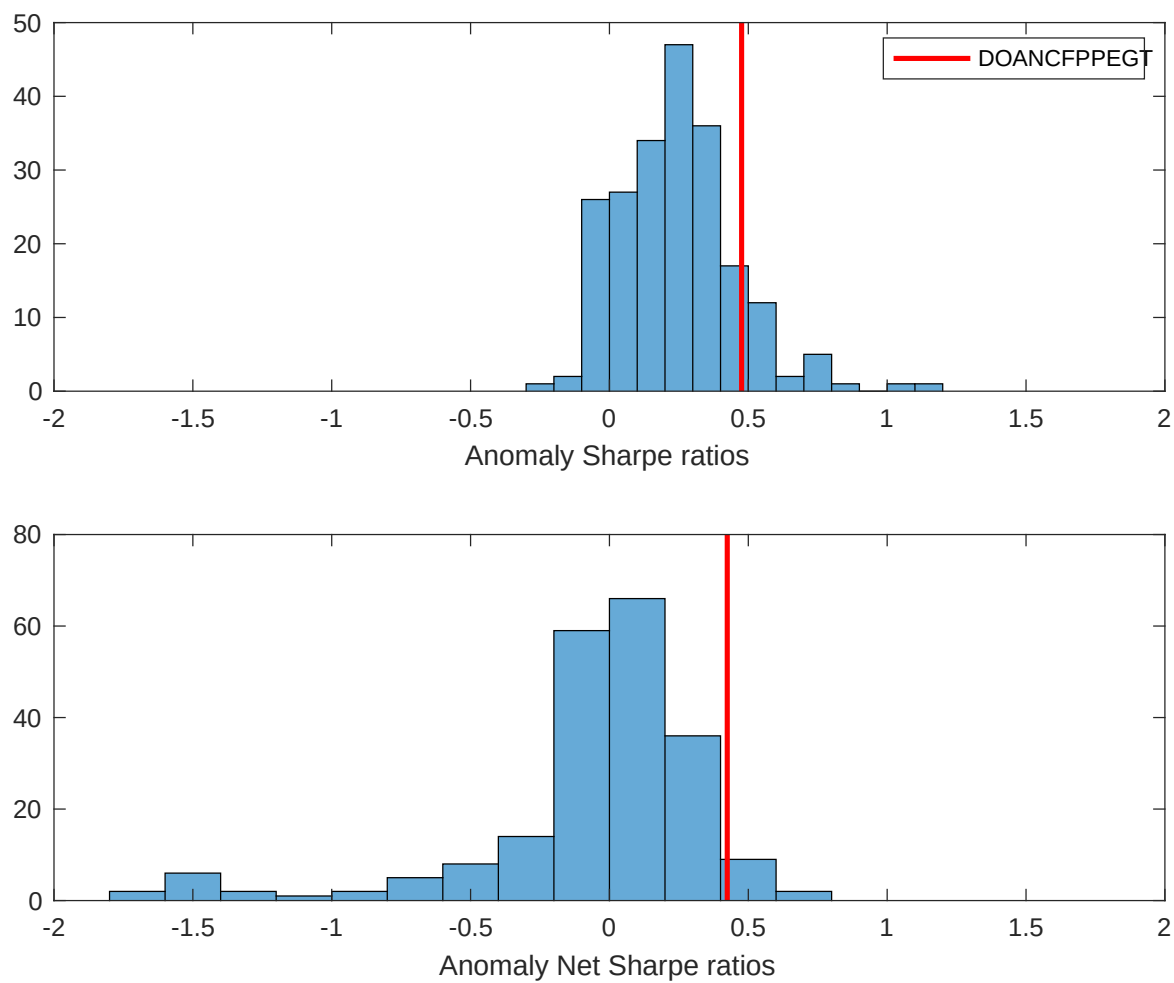


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the ACI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

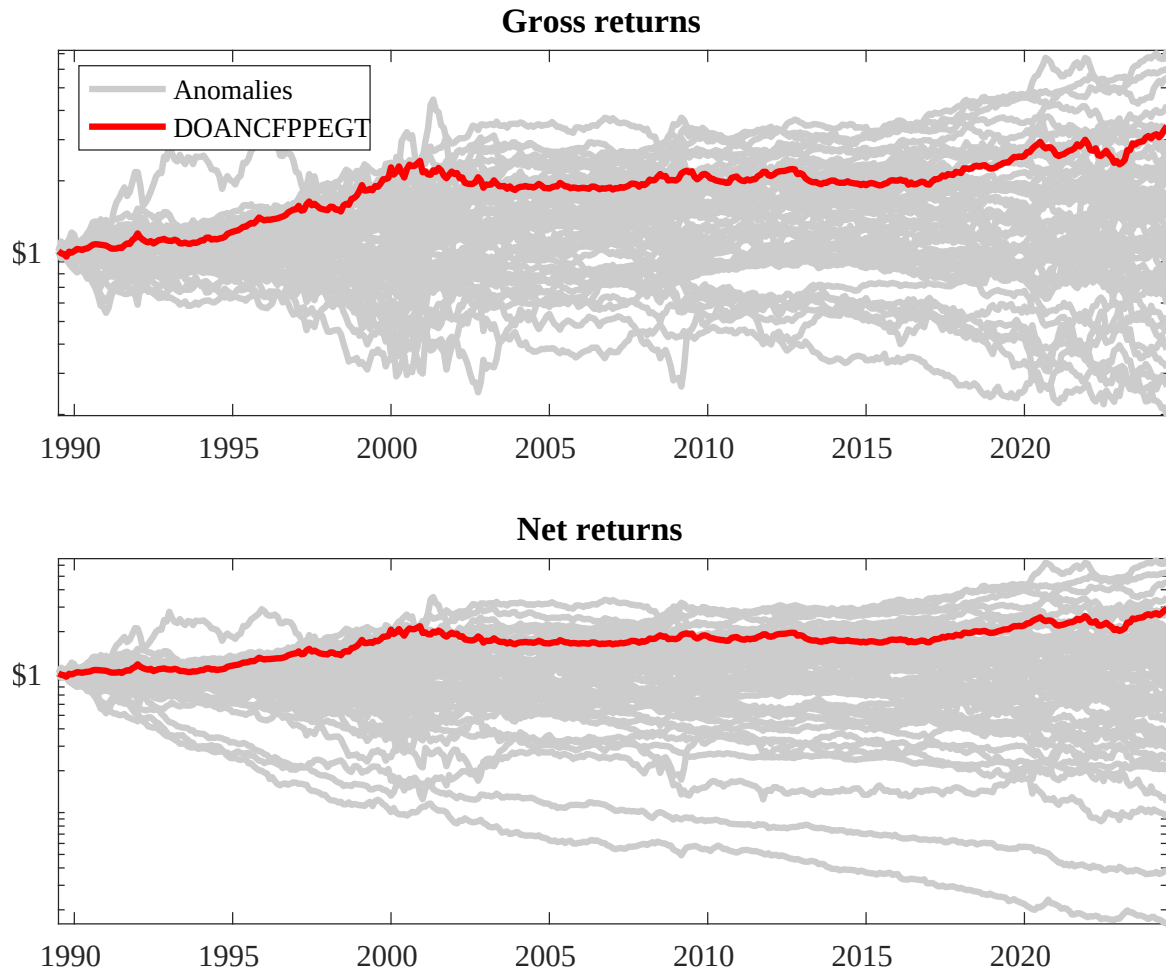
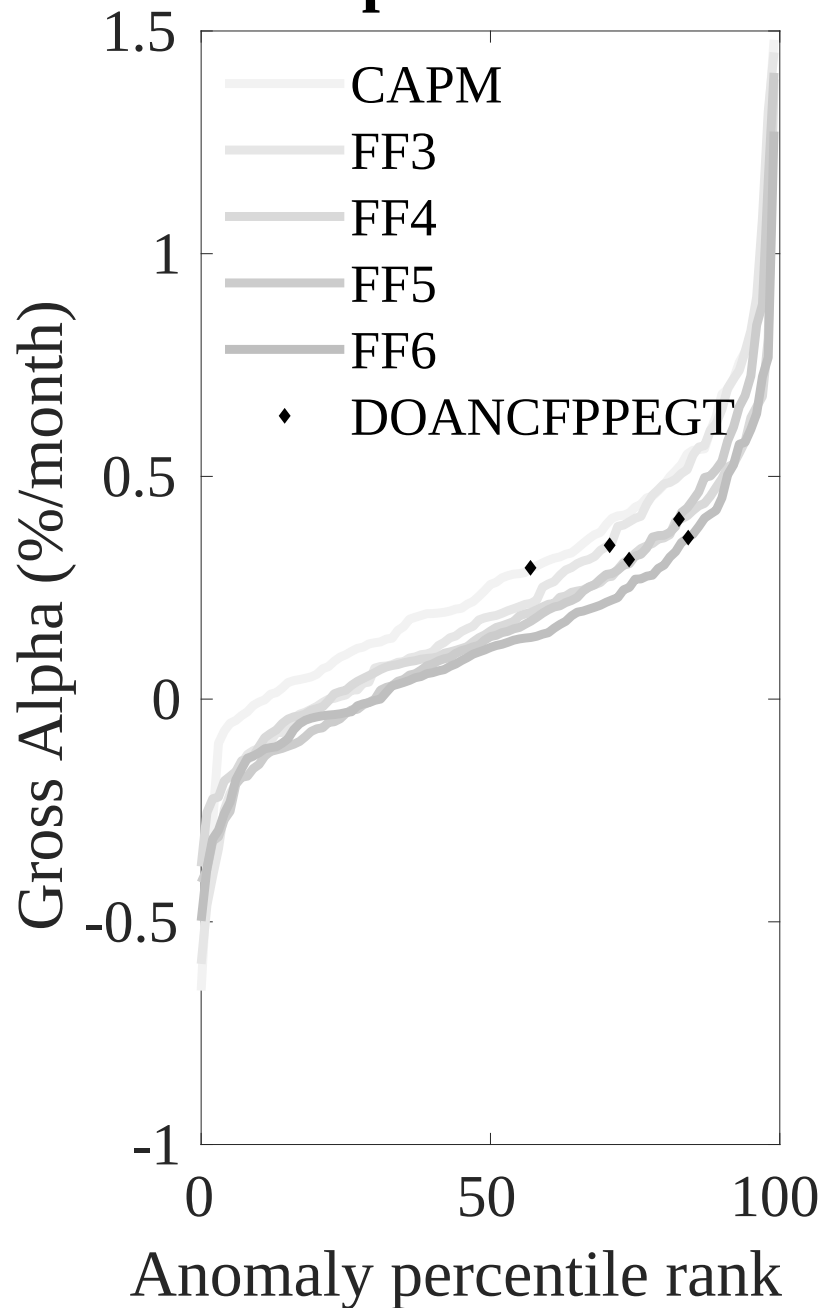


Figure 3: Dollar invested.

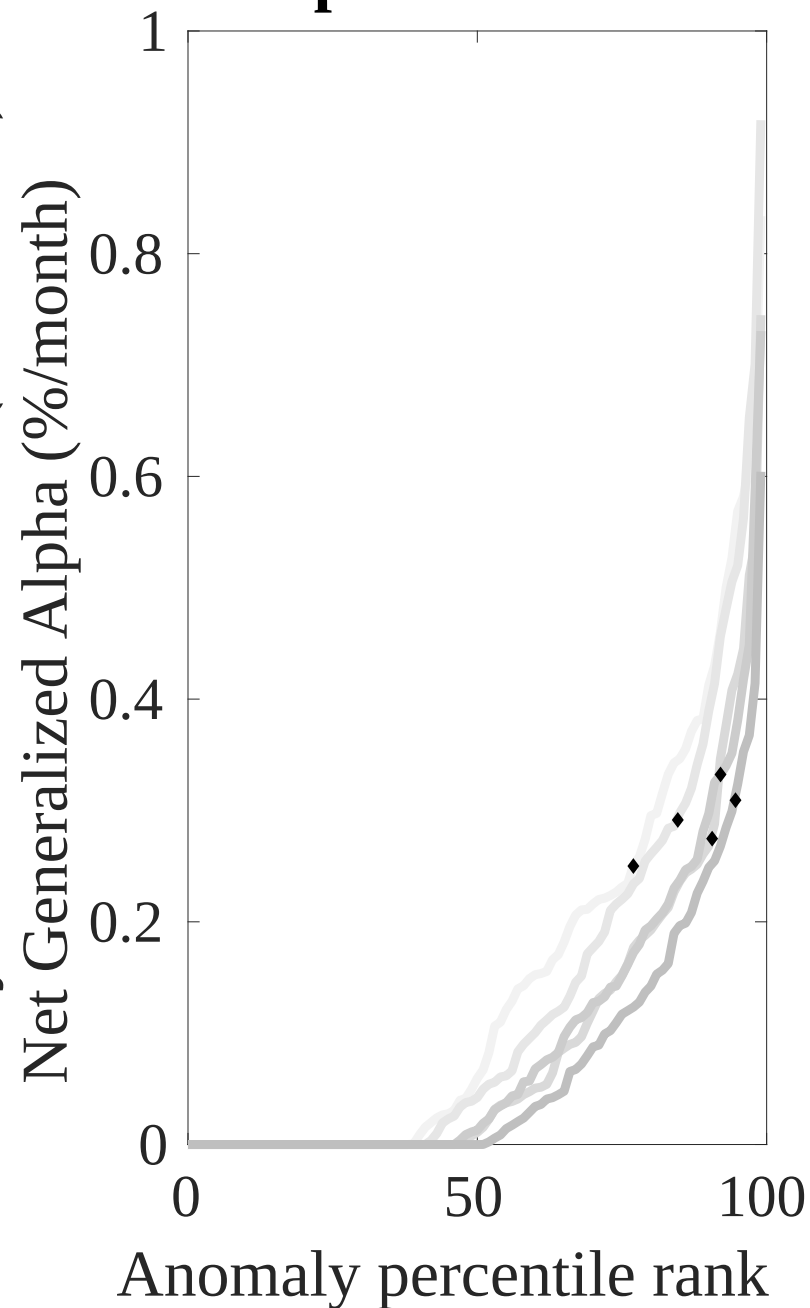
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the ACI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (RFS, 2016)



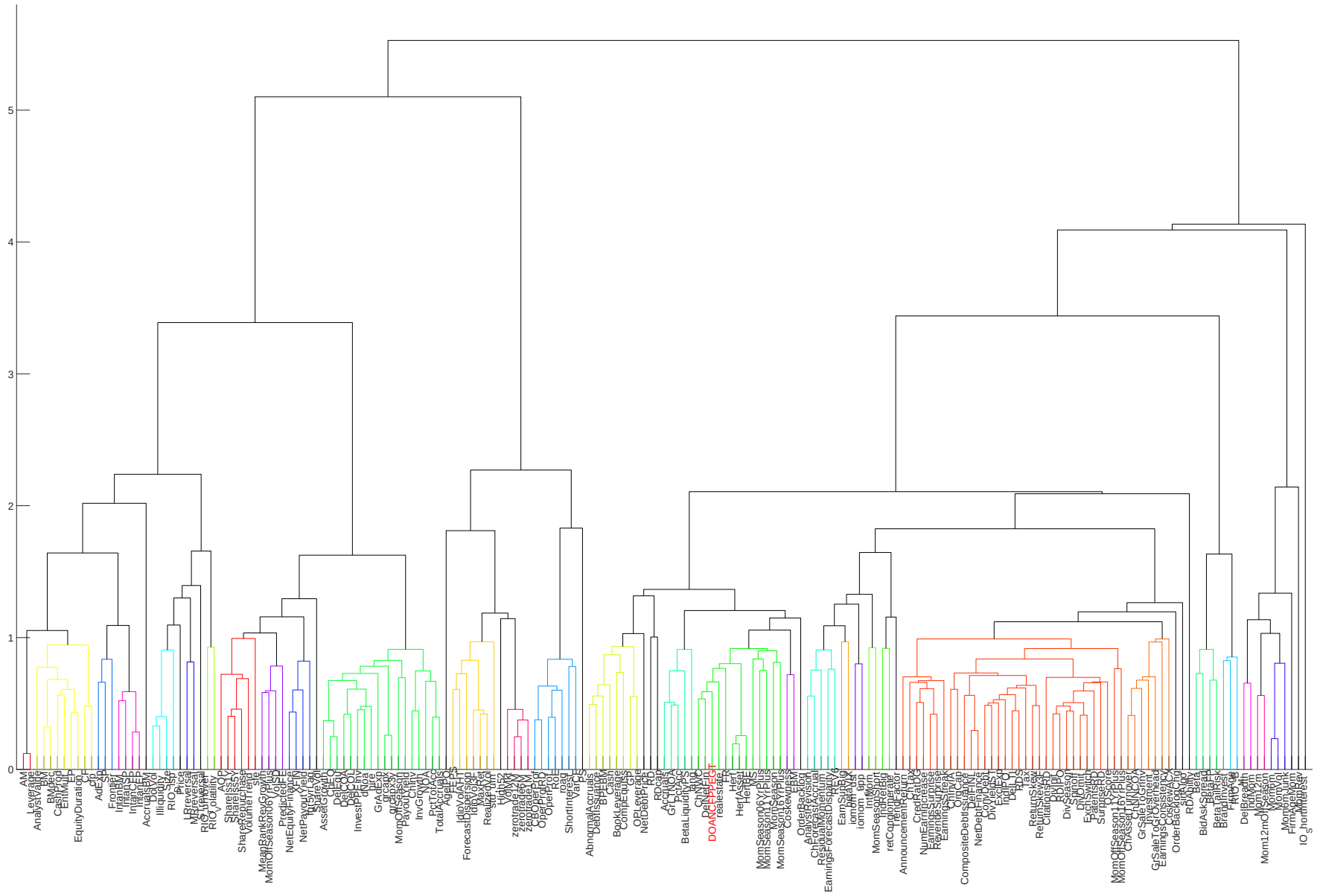


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

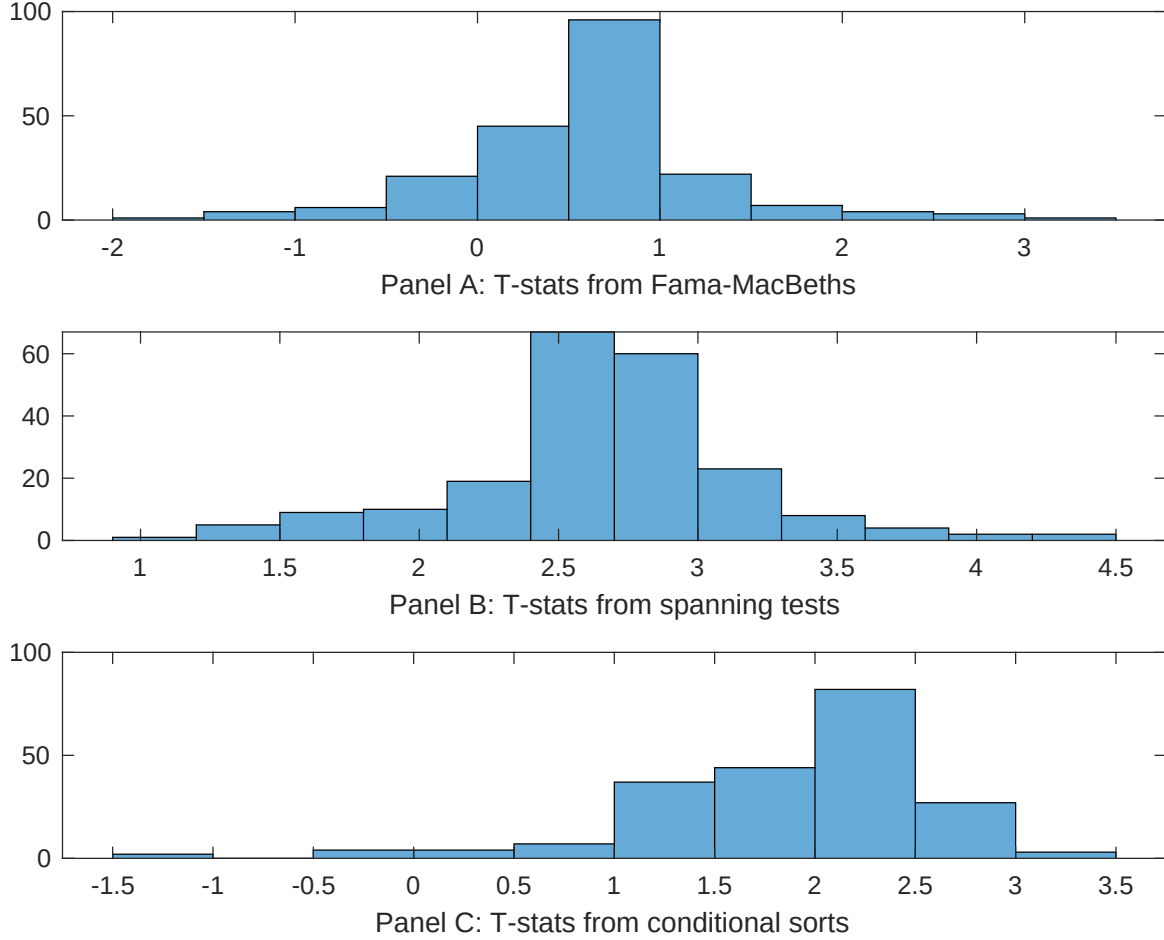


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of ACI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ACI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ACI}ACI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ACI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on ACI. Stocks are finally grouped into five ACI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ACI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on ACI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{ACI}ACI_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in equity to assets, Growth in book equity, Change in current operating liabilities, Abnormal Accruals, Operating Cash flows to price, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198906 to 202406.

Intercept	0.12 [3.97]	0.17 [5.65]	0.12 [3.94]	0.11 [3.84]	0.11 [3.64]	0.13 [4.28]	0.12 [4.53]
ACI	0.26 [1.26]	0.11 [0.51]	0.16 [0.78]	0.41 [0.17]	0.46 [0.00]	0.96 [0.47]	0.35 [0.17]
Anomaly 1	0.13 [4.51]						0.92 [2.13]
Anomaly 2		0.48 [6.27]					-0.25 [-0.30]
Anomaly 3			0.21 [4.41]				0.75 [1.40]
Anomaly 4				0.58 [0.19]			-0.92 [-0.27]
Anomaly 5					0.95 [3.11]		0.78 [2.42]
Anomaly 6						0.87 [7.58]	0.52 [5.03]
# months	420	420	420	420	415	420	415
$\bar{R}^2(\%)$	0	0	0	0	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the ACI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{ACI} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in equity to assets, Growth in book equity, Change in current operating liabilities, Abnormal Accruals, Operating Cash flows to price, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198906 to 202406.

Intercept	0.37 [3.75]	0.36 [3.64]	0.25 [2.64]	0.27 [2.79]	0.35 [3.43]	0.36 [3.72]	0.19 [2.00]
Anomaly 1	-18.31 [-3.62]						-11.83 [-1.38]
Anomaly 2		-7.96 [-1.48]					14.74 [1.82]
Anomaly 3			-34.74 [-7.99]				-26.85 [-5.60]
Anomaly 4				27.06 [6.08]			19.42 [4.29]
Anomaly 5					0.95 [0.24]		1.84 [0.50]
Anomaly 6						-22.95 [-4.46]	-9.36 [-1.40]
mkt	0.04 [0.02]	0.07 [0.03]	1.09 [0.47]	-2.51 [-1.03]	0.55 [0.22]	0.12 [0.05]	-1.01 [-0.44]
smb	-8.06 [-2.30]	-7.37 [-2.07]	-10.90 [-3.27]	-9.57 [-2.80]	-7.87 [-2.17]	-6.79 [-1.95]	-12.35 [-3.69]
hml	-8.60 [-2.00]	-9.78 [-2.25]	6.56 [1.44]	-6.83 [-1.63]	-10.61 [-2.07]	-8.60 [-2.02]	4.91 [0.95]
rmw	11.93 [2.69]	13.10 [2.92]	13.67 [3.27]	16.35 [3.76]	12.09 [2.59]	12.39 [2.82]	13.20 [3.06]
cma	-36.71 [-4.58]	-47.68 [-5.93]	-36.92 [-5.87]	-49.02 [-8.02]	-55.00 [-8.55]	-27.87 [-3.22]	-26.93 [-3.10]
umd	6.40 [2.96]	6.51 [2.96]	4.55 [2.22]	5.12 [2.42]	6.84 [2.42]	5.74 [2.67]	4.43 [1.69]
# months	420	420	420	420	416	420	416
$\bar{R}^2(\%)$	38	36	45	41	36	39	47

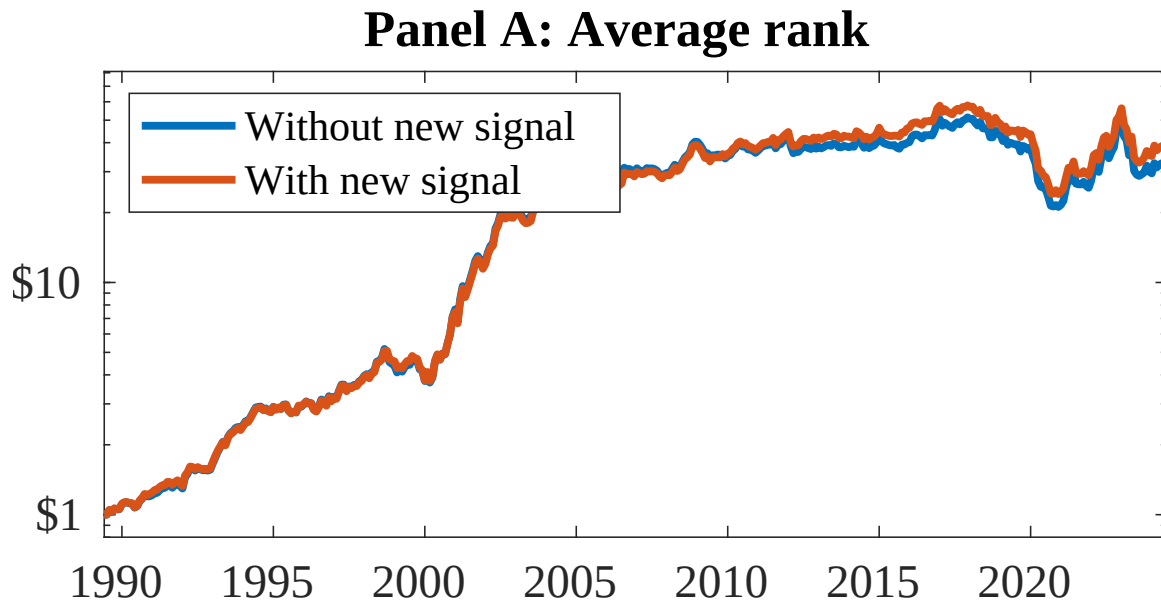


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as ACI. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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